

25 C For coil to remain horizontal,

$$\text{torque due to magnetic force} = mg \left(\frac{6.8 - 2.5}{100} \right)$$

$$BIL \left(\frac{15}{100} \right) = mg \left(\frac{6.8 - 2.5}{100} \right)$$

$$(60 \times 10^{-3})(I) \left(\frac{3.0}{100} \right) \left(\frac{15}{100} \right) = \left(\frac{5.0}{1000} \right) (9.81) \left(\frac{4.3}{100} \right)$$

$$I = 7.81 \text{ A}$$

26 D The coil is connected to an ammeter (a closed circuit), so there is induced current in the